

Box Problem

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Written report on general proof behind the lower bound strategy for a uniform distribution.

n is the amount of boxes, and thus also the range of values that can be contained inside these boxes (eg. 100 boxes have money between 1-100 dollars, 10 boxes have money between 1-10)

b is the value for the lower bound chosen, (eg. a lower bound of 8 means you only choose higher than 8)

We can calculate the probability of winning using this equation.

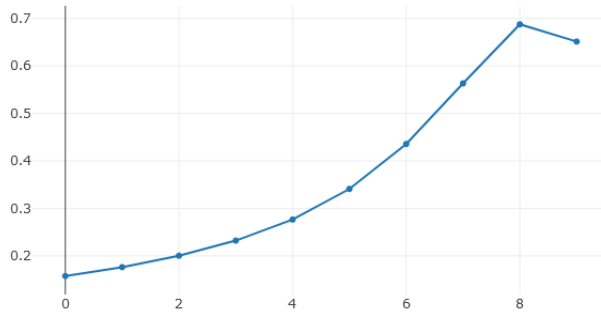
$$P(\text{win}) = \sum_{i=1}^n P(\text{win in box } i)$$

$$P(\text{win in box } i) = \sum_{j=0}^{(n-b)-1} P(\text{win in box } i \text{ with } n-j \text{ dollars})$$

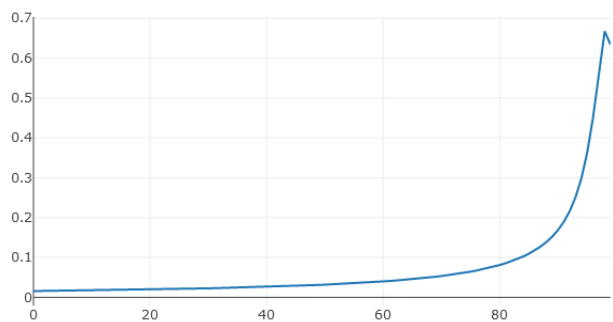
$$P(\text{win in box } i \text{ with } n-j \text{ dollars}) = \left(\frac{1}{n}\right) \cdot \left(\frac{b}{n}\right)^{i-1} \cdot \left(\frac{n-j}{n}\right)^{n-i}$$

$$P(\text{win}) = \sum_{i=1}^n \left[\sum_{j=0}^{(n-b)-1} \left(\frac{1}{n}\right) \cdot \left(\frac{b}{n}\right)^{i-1} \cdot \left(\frac{n-j}{n}\right)^{n-i} \right]$$

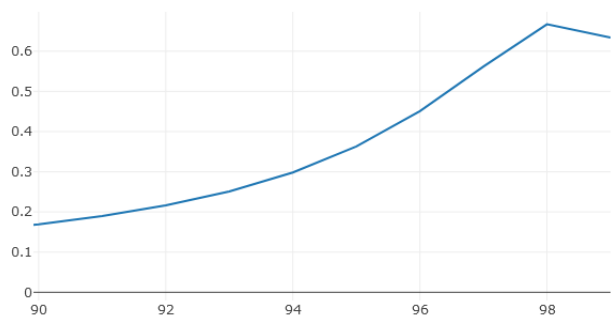
On the following plots, the y axis is the theoretical probability of winning, whereas the x axis accounts for the value of the lower boundary chosen (b) Here is a plot where $n = 10$.



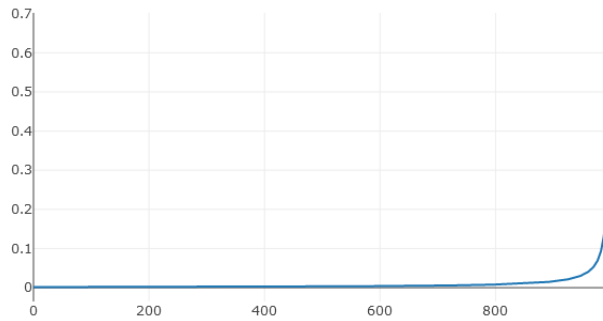
Here is a plot where $n = 100$



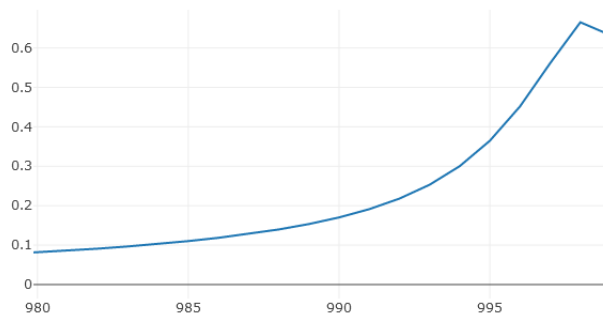
Here is the same plot, but the x axis (the lower boundary), only has values from 90 to 100



Here is a plot where $n = 1000$



Here is the same plot, but the x axis (the lower boundary), only has values from 980 to 1000



Observations:

- The highest chance is always between 0.65 and 0.70
- The highest probability is always $n - 2$, with $n-1$ being slightly lower.
- The function describes a gentle slope